PHY202: Physics II: Exam 6
Dr. Deskins

Excellent

$\frac{9}{9} \Rightarrow 100\%$

0 0 | 6 0.5 | 11 1 | 17 1.5 | 22 2 | 28 2.5 | 33 3 | 39 3.5 | 44 4 | 50 4.5 | 56 5 | 61 5.5 | 67 6 | 72 6.5 | 78 7 | 83 7.5 | 89 8 | 94 8.5 | 100 9

Name:_

These questions assess your understanding of the material from Chapters 25, 26 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. A scientist at a natural-history museum is holding a specimen 4.0cm (do) from a thin convex magnifying glass of focal length 9.6cm (f). What magnification (m) is produced?

$\frac{1}{do} + \frac{1}{di} = \frac{1}{f}$ $\quad \frac{1}{0.04} + \frac{1}{di} = \frac{1}{0.096}$ $\quad di = -0.069$

$-\frac{di}{do} = m$ $\quad m = -\frac{-0.069}{0.04} = +1.71$

ANSWER: +1.71 ✓

2. Calculate the critical angle for light traveling in a material that is surrounded by air (1.00), given that the material has an index of refraction of 1.3. ✓

$\theta_c = \sin^{-1}\left(\frac{n_2}{n_1}\right)$

$\theta_c = \sin^{-1}\left(\frac{1}{1.3}\right)$

$\theta_c = 50.28^\circ$

ANSWER: 50.28° ✓

3. What power of spectacle lens is needed to correct the vision of a near-sighted person whose far point (point) is 23. cm? Assume the spectacle lens is held 1.6cm (lens) away from the eye by eyeglass frames.

$di = -(\text{point} - \text{lens})$

$do = \infty = 0$

$di = -(0.23 - 0.016) = -0.214$

$\frac{1}{do} + \frac{1}{di} = \frac{1}{f}$

$0 + \frac{1}{-0.214} = \frac{1}{f}$

$P = \frac{1}{f} = -4.67\ D$

ANSWER: -4.67 D ✓

4. Calculate the speed at which light travels in a particular material given that the index of refraction of the material is 1.2 (n).

$n = \frac{c}{v}$

$1.2 = \frac{3.0 \times 10^8}{v}$

$v = 2.50 \times 10^8$ m/s

ANSWER: 2.50×10^8 m/s ✓

+4

5. Calculate the power of the eye when viewing an object 43. cm away. Take the lens-to-retina distance to be 2.5 cm.

P d_o d_i

ANSWER: 42.33 D ✓

$$\frac{1}{d_o}+\frac{1}{d_i}=\frac{1}{f} \quad \frac{1}{0.43}+\frac{1}{0.025}=\frac{1}{f}$$

$$P=\frac{1}{f}=42.33\,D.$$

6. A clear-glass light bulb is placed 0.68 m from a thin convex lens that has a focal length of 0.087 m. Calculate the distance from the lens to the location of the produced image.

d_o f d_i

ANSWER: 0.0998m ✓

$$\frac{1}{d_o}+\frac{1}{d_i}=\frac{1}{f} \quad \frac{1}{0.68}+\frac{1}{d_i}=\frac{1}{0.087}$$

$$d_i=0.0998\text{ m}$$

7. You shine a laser through air at an incident angle of 68.° into an unknown material and measure the angle of refraction to be 32.°. What is the unknown material's index of refraction?

$1.00 n_1$ θ_1 θ_2 n_2

ANSWER: 1.75 ✓

$$n_1\sin\theta_1=n_2\sin\theta_2$$
$$1.00\sin(68)=n_2\sin(32)$$
$$0.9272=n_2\sin(32)$$
$$0.9272=n_2(0.5299)$$
$$n_2=1.75$$

8. What is the size of the image on the retina of a 2.5 mm-diameter strand of hair that is held at arm's length (43. cm) away? Take the lens-to-retina distance to be 2.1 cm.

h_i h_o d_o d_i

ANSWER: -1.22×10^{-4}m ✓

$$\frac{h_i}{h_o}=-\frac{d_i}{d_o} \quad \frac{h_i}{2.5\times10^{-3}}=-\frac{0.021}{0.43}$$

$$\frac{h_i}{1.5\times10^{-3}}=-0.0488$$

$$h_i=-1.22\times10^{-4}\text{ m}$$

+4

9. Calculate the power of the eye when viewing an object 7.7 km away. Take the lens-to-retina distance to be 2.1 cm.

P, do, di

ANSWER: 47.62 D ✓

$$\frac{1}{do} + \frac{1}{di} = \frac{1}{f}$$

$$\frac{1}{7.7\times10^3} + \frac{1}{0.021} = \frac{1}{f}$$

$$P = \frac{1}{f} = 47.62\ D$$

+1

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1. $m = 1.71 \because \frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f}, \frac{h_i}{h_0} = -\frac{d_i}{d_0} = m$
2. $\theta_c = 50.3° \because n_1 \sin\theta_1 = n_2 \sin\theta_2$
3. $P = -4.67\,\mathrm{D} \because \frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f}, P = \frac{1}{f}$
4. $v = 2.50 \times 10^8\,\frac{\mathrm{m}}{\mathrm{s}} \because n = \frac{c}{v}$
5. $P = 42.3\,\mathrm{D} \because \frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f}, P = \frac{1}{f}$
6. $d_i = 0.0998\,\mathrm{m} \because \frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f}$
7. $n_2 = 1.75 \because n_1 \sin\theta_1 = n_2 \sin\theta_2$
8. $h_i = -1.22 \times 10^{-4}\,\mathrm{m} \because \frac{h_i}{h_0} = -\frac{d_i}{d_0} = m$
9. $P = 47.6\,\mathrm{D} \because \frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f}, P = \frac{1}{f}$